PART II - INDUSTRY LENS | CHAPTER 04

China's AI stack localizes under constraint

Emerging signal | Evidence current to 27 August 2026 | Medium on localization direction; low-medium on the cited 2030-35 market forecasts

Research question

How does China's AI stack localizes under constraint change enterprise economics, competitive advantage and executive priorities?

Executive Summary

China's AI stack is localizing under constraint. Compute, memory, advanced packaging, model efficiency, cloud delivery, power and policy are increasingly interdependent because weakness in one layer limits the usefulness of progress in another. The near-term signal is clear: companies and policymakers are investing across the stack rather than relying on a single substitute. What remains unproven is whether announced capacity can achieve competitive yield, energy efficiency, software compatibility, utilization and customer economics at scale. [1][4]

Evidence Boundary and Confidence: This is an emerging signal. The quantitative forecasts come from one Daily Brief citing external research and require continued verification; ten Daily editions cannot establish a structural outcome by themselves.

Quantitative anchors

FORECAST **92% to 34%** forecast reduction in China's supply-demand gap for logic chips at 7 nm and below between 2025 and 2035, cited from Goldman research in the Daily Brief. [1]	FORECAST **46% CAGR** forecast growth in China's advanced-process wafer supply from 2025 to 2035, reaching 410,000 wafers per month. [1]
FORECAST **RMB 4.56 trillion** forecast 2030 total addressable market for AI chips in China cited in the same Daily Brief; inference was estimated at about 70%. [1]	FORECAST **~70%** forecast share of China's 2030 AI-chip market attributed to inference in the cited Daily Brief; this is a forecast from one short-window source. [1]

Note: each card was checked against the cited passage; measurement type is shown above the value.

Key Findings

01	**Constraint links the stack** Restricted access to one layer increases the value of substitutes in adjacent layers. [1][4]
02	**Efficiency can offset hardware gaps** Model design, quantization, workload orchestration and software optimization can reduce dependence on the most advanced hardware, although performance and developer costs must be measured in real applications. [7][1]
03	**Policy accelerates capacity but not necessarily returns** Public support and strategic demand can fund rapid build-out. [6][8]

Evidence and Evolution, 2023-2026

MULTI-YEAR CONTEXT

Technology fragments

The CEO archive frames export controls, investment screening and geopolitical risk as persistent strategic variables. Technology access becomes part of resilience planning. [2][3]

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2025-EARLY 2026

Efficiency and local ecosystems matter more

External digital and trade research points to continued technology diffusion alongside fragmented standards, infrastructure and supply chains. [4][5]

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AUGUST 2026

The Daily feed shows stack-wide activity

Daily items cover chips, memory, packaging, cloud, models and financing. The breadth supports a localization hypothesis, but the time window is too short to judge sustainable market share or returns. [6][7][1][8]

Figure 1. Evolution of the evidence base

Note: stages summarize the sequence in the archive; they do not imply a uniform adoption rate across markets or companies.

Analytical Architecture

01 Constraint links the stack	02 Efficiency can offset hardware gaps	03 Policy accelerates capacity but not necessarily returns
Restricted access to one layer increases the value of substitutes in adjacent layers. Domestic compute is less useful without memory, packaging, software and power that can deliver a reliable workload. [1][4]	Model design, quantization, workload orchestration and software optimization can reduce dependence on the most advanced hardware, although performance and developer costs must be measured in real applications. [7][1]	Public support and strategic demand can fund rapid build-out. Commercial success still depends on yield, utilization, ecosystem compatibility and customer willingness to pay. [6][8]

Figure 2. Causal architecture of the finding

Source: Weekly Brief Atlas analysis of the cited Briefs and authoritative research.

Economic Assessment

The stack should be evaluated through delivered workload economics rather than nominal chip specifications. Relevant measures include yield, energy per useful task, software migration cost, developer productivity, utilization and total cost of ownership. Forecast reductions in supply gaps can coexist with weak economics if capacity is expensive or incompatible with customer workloads. Global companies also face an option problem: supporting multiple stacks raises engineering cost but can preserve market access and resilience. [1][4][9]

Implications

Chinese technology companies	Winning requires coordinated progress across hardware, software, cloud and applications rather than isolated capacity announcements.
Global vendors	Products and roadmaps need explicit scenarios for export controls, local standards and multi-stack support.
Investors	Separate strategic importance from commercial return and demand proof on yield, utilization, energy efficiency and customer revenue.

Figure 3. Executive implication matrix

Recommendations

01 NEXT 90 DAYS **Map stack dependencies** Identify critical dependencies across compute, memory, packaging, software, cloud, power and policy for each product line.	**02 NEXT 90 DAYS** **Build scenario roadmaps** Define product and sourcing choices under narrow, broader and easing restriction cases.
03 6-12 MONTHS **Test multi-stack economics** Measure migration cost, performance, energy and developer productivity on representative workloads.	**04 6-12 MONTHS** **Verify commercial utilization** Track revenue-bearing deployments and customer retention separately from announced capacity.

Figure 4. Sequenced management response

Conditions That Would Weaken the Finding

- Restrictions could ease or narrow, reducing the economic value of some localization investments.
- Domestic volume may rise without closing performance, yield or ecosystem gaps.
- Forecast market size may overstate monetizable demand if inference prices fall rapidly.

Monitoring Framework

01	Advanced-process yield and energy efficiency
02	Revenue-bearing utilization
03	Software migration cost
04	Domestic versus imported dependency by layer
05	Scope of export and investment controls

Evidence Boundary and Confidence

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Notes and Sources

Superscript numbers in the chapter refer to this list. Page references use the printed report pagination where available.

[1] Greater China TMT Daily Intelligence Brief · 2026-08-26 · AI investments continue from Agent applications to chips, data and infrastructure

[2] Weekly Brief · 2024-06-24 · How to Face Rising Geopolitical Risk

[3] Weekly Brief · 2025-01-14 · Navigating the New Dynamics of Global Trade

[4] Authoritative research · OECD · 2024-11-19 · OECD Digital Economy Outlook 2024 (Volume 2): Strengthening Connectivity, Innovation and Trust

[5] Authoritative research · WTO · 2025-12-15 · Global Value Chain Development Report 2025: Rewiring GVCs in a Changing Global Economy

[6] Greater China TMT Daily Intelligence Brief · 2026-08-10 · Capital, supply chain and energy constraints of AI algorithms and semiconductor expansion

[7] Greater China TMT Daily Intelligence Brief · 2026-08-17 · The United States has tightened AI's cooperation with chips, and the financing of computing and data centre construction continues to expand.

[8] Greater China TMT Daily Intelligence Brief · 2026-08-27 · Model drop-down and domestic capacity validation speed, AI capital cycle entering return review phase

[9] Weekly Brief · 2026-07-28 · The AI Bill Lands on the CEO’s Desk

[10] Greater China TMT Daily Intelligence Brief · 2026-08-13 · AI and cloud data centre driven infrastructure and semiconductor supply and demand re-engineering

[11] Greater China TMT Daily Intelligence Brief · 2026-08-11 · Capacity and compliance options under AI algorithms and semiconductor supply chain pressure

[12] Authoritative research · ITU · 2025-11-27 · Global Connectivity Report 2025

Independent analysis of the available Weekly Brief archive and selected authoritative research. The chapter distinguishes reported actuals, forecasts, scenarios, surveys, modelled estimates and company-reported figures. Figures are not presented as audited actuals unless the source supports that characterization.